CStats - Lab #2

March 3, 2026

Exercise 1.
Lake Macatawa, an inlet lake on the east side of Lake Michigan, is divided into an east basin and a west basin. To determine whether the bacteria count was lower in the west basin of Lake Macatawa than in the east basin, $n = 37$ samples of water were taken from the west basin and the number of bacteria colonies in 100 milliliters of water was counted. The sample characteristics were $\bar{x} = 11.95$ and $s = 11.80$, measured in hundreds of colonies. Find the approximate 95% confidence interval for the mean number of colonies (say, μ_w) in 100 milliliters of water in the west basin.

Exercise 2.
A light bulb manufacturer claims that the average lifespan of its bulbs is 170 hours. To verify this claim, a consumer protection agency randomly selects 100 bulbs from a production batch. After testing, they find an average lifespan of 158 hours with an empirical standard deviation of 30 hours. Assuming the lifespan follows a normal distribution, can we conclude from this survey that the manufacturer's claim is false?

Exercise 3.
A random sample of $n = 9$ wheels of cheese yielded the following weights in pounds, assumed to be $\mathcal{N}(\mu, \sigma^2)$:

21.50	18.95	18.55	19.40	19.15
22.35	22.90	22.20	23.10	

(a) Give a point estimate of σ.

(b) Find a 95% confidence interval for σ.

(c) Find a 90% confidence interval for σ.

Exercise 4.
To verify the regularity of bus schedules, a bus company conducted 71 recordings of time gaps (in seconds) between bus arrivals at a stop and the observed average schedule. The unbiased estimate of the parameter σ^2 obtained was 5100. Deduce a 95% confidence interval for this measure of schedule regularity.

Exercise 5.

To estimate the accuracy of a thermometer, $n = 100$ independent measurements are taken of a liquid maintained at a constant temperature of 20 degrees Celsius. The observations $x_i,\ 1 \le i \le 100$, produced the value:

$$\sum_{i=1}^{100} x_i^2 = 40011,$$

Construct a 95% confidence interval for the precision of this thermometer, measured by the variance σ^2 of the measurements.